MILITARY SPECIFICATION

POWER SUPPLY: 10527250

1. SCOPE

1.1 This specification covers one type of transistorized power supply capable of supplying regulated voltages, both alternating current (ac) and direct current (dc). It performs this function only in conjunction with several transistor circuits that are not integral with this unit.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communication, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F10527250	Power Supply: 10527250

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PACKAGING DATA SHEET

MIL-P-14232/P10527250 Packaging of Power Supply: 10527250

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The power supply shall be fabricated in accordance with Drawing F10527250 and drawings pertaining thereto.

3.2.1 Treatment, moisture-and fungus-resistant.- The power supply shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship
(e) Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperature.- The power supply shall meet the performance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to ambient temperatures at -80° and at +160°F.

3.4.2 Operating temperature.- The power supply shall meet the performance requirements of 3.5 while exposed to temperatures at -40° and at +150°F.

3.4.3 Altitude.- The power supply shall meet the performance requirements of 3.5 at an altitude of 30,000 feet above sea level.

3.4.4 Shock.- The power supply shall withstand impact shocks up to 15 G's; each shock impulse having a time duration of 11 ± 1 milliseconds (ms). Subsequently, the power supply shall meet the performance requirements of 3.5.

3.4.5 Vibration.- The power supply shall meet the performance requirements of 3.5 after being subjected to vibrations up to 5 G's over a frequency range of 5 to 500 cycles per second (cps).

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3.5 Performance.- The power supply consists of two independently functioning regulators. One is used for the ± 26 volt direct current (Vdc) supply and the other is used for the 400 cps supply. In order to verify that each regulator performs its function, the power supply must be connected to a simulated circuit representing the regulated load.

3.5.1 Regulator No. 1 characteristics.

3.5.1.1 Inputs.- The regulator shall accept three input voltages:

(a) +28 Vdc ± 10%
(b) 26 volt alternating current (Vac) root mean square (rms), two-phase supply
(c) a 24 Vdc signal that is being regulated

3.5.1.2 Output regulation.- The output shall be regulated within ± 5% about the nominal value of 24 Vdc when loaded by a 2 ampere resistive load for input voltage variations from 25 Vdc to 31 Vdc. Below the 25 Vdc input level, the regulator output shall linearly follow the input down to 15 Vdc.

3.5.1.3 Output impedance.- The output impedance of the regulated output shall be less than 0.8 ohm for loads up to 5 amperes.

3.5.1.4 Output voltage adjustment.- The regulator output shall be adjustable to maintain a 25V output.

3.5.1.5 Short-circuit protection.- The regulator shall have short circuit protection such that if the regulated output is short circuited, neither the regulator circuitry nor the external circuitry shall be damaged. The threshold for activation of short-circuit protection circuits shall be 7.0 ± 2.0 amperes.

3.5.1.6 AC ripple.- The dc regulated voltage at the loaded output shall have less than 2% ripple for a 1-ampere load and shall have a peak ripple voltage less than 3 volts.

3.5.2 Regulator No. 2 characteristics.

3.5.2.1 Input voltage.- The regulator shall accept 28 Vdc ± 10%.

3.5.2.2 Output regulation.- The regulated output, suitably loaded, shall have a regulation of ± 2.5% about a nominal 19.0 Vdc for output load currents from 100 to 400 milliamperes (ma) and for input variations from 25 Vdc to 31 Vdc.

3.5.2.3 Output impedance.- The regulator shall have an output impedance of 2.0 ohms ± 0% -20% under dynamic loads from 100 ma to 400 ma.

3.5.2.4 AC ripple.- The dc regulated output shall have less than 2% ripple at a 0.4 ampere load.

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3.5.2.5 Regulated output voltage adjustment.- The output voltage shall have an adjustment of ± 5% about a nominal voltage of 19 Vdc.

3.6 Radio interference suppression.- Interference control requirements shall be in accordance with MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE I

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Altitude	3.4.3	4.6.2.3
302. Shock	3.4.4	4.6.2.4
303. Vibration	3.4.5	4.6.2.5
304. Radio interference suppression	3.6	4.6.5

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.5 Percent Defective		
101. Moisture and fungus resistance	3.2.1	4.6.1
102. Input	3.5.1.1	4.6.3.2
103. Output regulation	3.5.1.2	4.6.3.3
104. Output impedance	3.5.1.3	4.6.3.4
105. Output voltage adjustment	3.5.1.4	4.6.3.5
106. Short-circuit protection	3.5.1.5	4.6.3.6
107. AC ripple	3.5.1.6	4.6.3.7
108. Input voltage	3.5.2.1	4.6.4.1
109. Output regulation	3.5.2.2	4.6.4.2
110. Output impedance	3.5.2.3	4.6.4.3
111. AC ripple	3.5.2.4	4.6.4.4
112. Regulated output voltage adjustment	3.5.2.5	4.6.4.5
MINOR: NONE		

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three units shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Fabrication	3.2	Applicable drawing - visual
402.	General specification	3.3	MIL-F-13926 - visual
403.	Temperature (storage)	3.4.1	4.6.2.1
404.	Temperature (operating)	3.4.2	4.6.2.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment

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will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus-resistant.- The power supply shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.2 Environmental.

4.6.2.1 Storage temperature.- The power supply, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, Procedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the power supply shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temperature.- The power supply shall be subjected to the Temperature Cycling Test, Procedure I of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the power supply shall meet the performance requirements specified in 3.5.

4.6.2.3 Altitude test.- The power supply shall be placed within a test chamber and the absolute internal pressure reduced to 8.88 inches of mercury (corresponding to an altitude of 30,000 feet above sea level). After sufficient time has elapsed to allow all entrapped air to escape, the power supply shall meet the requirements specified in 3.5.

4.6.2.4 Shock test.- The power supply shall be tested in accordance with MIL-F-13926, Procedure II. The power supply shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the power supply mounted and operating at room ambient conditions, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the power supply shall be closely inspected for any evidence of mechanical failure, loose or broken parts. At end of test, the power supply shall meet the performance requirements of 3.5.

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4.6.3 Performance.

4.6.3.1 Regulator No. 1.- In order to load the regulator of the power supply, connect the test circuit shown in Figure 1. Perform all tests with this test circuitry connected.

4.6.3.2 Input voltage test.- Using standard measuring equipment, verify that the input voltages to the regulator are in compliance with 3.5.1.1.

4.6.3.3 Regulation test.- Using the test circuit of Figure 1, load the output (pins H and C) with a 2.0 ampere resistive load. While varying the nominal 28Vdc source (pin N) between 25 and 31Vdc, measure the regulated output with standard measuring equipment for conformance with the regulation requirements of 3.5.1.2. Reduce the nominal 28V source below 25 volts and verify that regulator output conforms with 3.5.1.2.

4.6.3.4 Output impedance test.- With all inputs at their nominal values, measure the no-load voltage of the regulator output (pins H and C). Then, adjust the load for 5.0 amperes and measure the regulator output again. The difference of these two readings divided by 5.0 shall be equal to or less than 0.8 ohm in compliance with 3.5.1.3.

4.6.3.5 Output voltage adjustment.- With all inputs set at their nominal values, and with a 2 ampere load on the output (pins H and C), measure the output while varying the adjustment (R9) across its limits for compliance with 3.5.1.4.

4.6.3.6 Short-circuit protection.- Place a short-circuit of less than 0.5 ohm across the output pins H and C for 5 seconds. Observe that the output voltage is reduced to less than 5 volts. After removal of the short-circuit, the output shall return to normal.

4.6.3.7 AC ripple.- With all inputs at their nominal value regulator output loaded by a 1.0 ampere load and using standard measuring equipment, measure the ac ripple across the load (pins H and C) to insure compliance with 3.5.1.6.

4.6.4 Regulator No. 2.- Load the regulator with the test circuit of Figure 1. Perform all tests using test circuit.

4.6.4.1 Input voltage test.- Using standard measuring equipment, verify that the input voltage to the regulator is in compliance with 3.5.2.1.

4.6.4.2 Regulation test.- Load the output with a 400 ma load. While varying the nominal 28Vdc source (pin N) between 25 and 31Vdc, measure the regulated output (pins S and C) using standard measuring equipment. Repeat for a 100 ma load. Regulation shall conform with 3.5.2.2.

4.6.4.3 Output impedance test.- With all inputs at their nominal values, measure the output voltage (pins S and C) for 100 ma and 400 ma load current. The difference between the load voltages divided by .300 amp shall be equal to or less than 2.0 ohms in compliance with 3.5.2.3.

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4.6.4.4 AC ripple.- With all inputs at their nominal value regulator output (pins S and C) loaded by a 0.4 ampere load, and using standard measuring equipment, measure the ac ripple across the load for compliance with 3.5.2.4.

4.6.4.5 Output voltage adjustment.- With all inputs at their nominal values, and with a 0.4 ampere resistive load, measure the output (pins S and C) while varying the adjustment (R18) across its limits for compliance with 3.5.2.5.

4.6.5 Radio interference test.- The power supply shall be tested in accordance with MIL-I-6181 to insure conformance with 3.5.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527250. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Power Supply 10527250 in conjunction with power devices located in the Amplifier Electronic Control 10527247 provides regulated dc and ac voltages for stabilization electronic operation of the Antioscillation Sighting System XM58. It also provides overload protection for the electronic control amplifier.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing (see 5.1)
c. Applicable packaging data sheet number (see 5.1)
d. Applicable stock number

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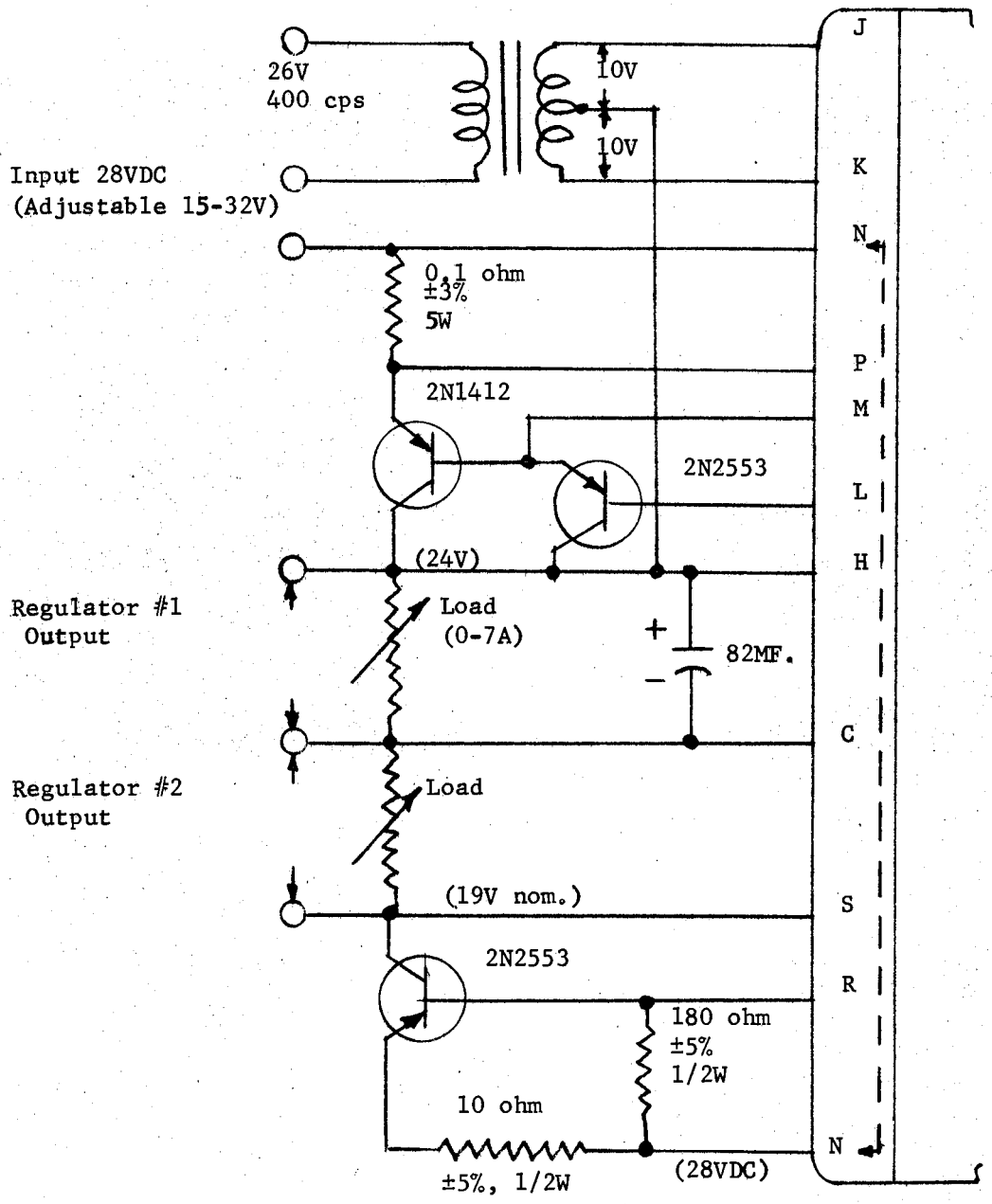


Figure 1. Test Circuit for Power Supply: 10527250

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